

No. of Functional Features Included in the Solution

Date	15 March 2026
Team ID	xxxxxx
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Audio Upload	Allows users to upload a voice recording explaining a concept	Audio Input Module	Done	1
2	Speech-to-Text Transcription	Transcribes uploaded speech into text using OpenAI Whisper	Transcription Module	Done	1
3	Semantic Similarity Scoring	Compares transcribed explanation with a reference concept using Sentence-BERT embeddings	NLP / Scoring Module	Done	1
4	Filler Word & Pause Detection	Detects filler words (um, like, uh), pause ratio, and RMS energy using Librosa	Audio Analysis Module	Done	0.5
5	Conceptual Understanding Score	Generates a comprehension score with qualitative feedback (Strong / Moderate / Poor Understanding)	Scoring Engine	Done	1
6	Waveform Visualization	Displays the audio waveform of the recorded/uploaded explanation	Visualization Module	Done	0.5
7	Interactive Streamlit Dashboard	Displays transcription, scores, filler word stats, and pause analysis in one dashboard	UI / Dashboard Module	Done	0.5
8	PDF Report Generation	Generates a downloadable PDF report with waveform images, metrics, and summaries	Reporting Module	Done	0.5

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5
Additional / Nice-to-Have Features	3
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Streamlit Dashboard, Waveform Visualization Panel

S.No	Category	Features in Category	Example Features
2	Backend / Logic	3	Scoring Engine, Semantic Similarity Module, Fluency Analysis Module
3	Database / Storage	1	Temporary audio file & session-state storage
4	API / Integration	1	OpenAI Whisper model integration
5	Security / Authentication	0	Not applicable (no user login required)